Task-Specific Heterogeneous Network for Object Detection in Aerial Images

Ying Yu, Xi Yang, *Member, IEEE*, Jie Li, and Xinbo Gao, *Senior Member, IEEE*

***Abstract*— Object detection in aerial images has attracted increasing attention in recent years. Due to the complex background and arbitrary-oriented objects, it is challenging to accurately locate the objects of interest in the images. Many methods have been developed for improving localization accuracy of oriented objects. However, classification and localization tasks require different features due to the unique characteristics of aerial images, which are still not fully considered in previous methods. Therefore, we propose a task-specific heterogeneous (TSH) network for aerial object detection. Specifically, we design an interference-suppression module (ISM) to reduce both the background and interclass interference, which can provide discriminative features for classification. To produce more reliable localization confidence, we propose a joint-learning quality estimation (JQE) module to adaptively combine the classification and regression features, thereby achieving accurate classification and localization quality estimation simultaneously. Moreover, we propose a point-based localization (PBL) branch. In the PBL, the learnable points can effectively adapt to objects with diverse shapes and orientations, and the dynamic information aggregation (DIA) module can enhance the relationships between the dispersible points to promote localization accuracy. The proposed TSH is evaluated extensively on four widely used aerial datasets, demonstrating its state-of-the-art performance. Ablation study and visualizations further verify the effectiveness of our method.**

***Index Terms*— Aerial images, heterogeneous network, object detection.**

I. Introduction

OBJECT detection in aerial images is a fundamental and challenging task, involving predicting accurate bounding boxes and corresponding categories of objects within an image [1], [2], [3], [4], [5]. This task plays a vital role in a variety of applications, such as port management [6], traffic surveillance [7], maritime rescue [8], [9], and urban planning [10]. Meanwhile, it faces challenges due to the arbitrary directions and scales of objects, along with complex backgrounds.

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In practice, the classification and localization tasks have different objectives. The classification task focuses on extracting effective semantic information to assign proper category labels to the samples, while the localization task emphasizes obtaining detailed boundary information to accurately locate the objects. Considering the difficulty of capturing arbitrary-oriented objects, the existing aerial object detection methods are devoted to improving the localization accuracy of oriented bounding boxes (OBBs) from different perspectives. For instance, in view of feature calibration, single-shot alignment network (S^2ANet) [11] extracts rotation-invariant features via an alignment convolution to enhance subsequent detection. Regarding regression optimization, gliding vertex [5] and circular smooth label (CSL) [12] construct various oriented object representations to mitigate the limitations of angle prediction. Furthermore, rotation sensitive detector (RSDet) [13] and Gaussian wasserstein distance (GWD) [14] design effective loss functions to facilitate the learning of different OBB representations. As for label assignment, General Gaussian Heatmap Label Assignment (GGHL) [15] and detector with feature enhancement and soft label assignment (FSDet) [16] propose more suitable strategies for oriented objects, accommodating diverse shapes and orientations. These methods have achieved promising results and considerable progress in detecting oriented objects.

However, the distinct requirements of classification and localization tasks in aerial images have not been fully considered. Specifically, for classification, there may exist a large amount of background noise and mutual interference among classes in aerial images, making it difficult to learn discriminative features for object recognition. Moreover, when the structure of classification and regression is excessively decoupled, the estimated classification confidence may not align with the real localization quality, as shown in Fig. 1(a). As for regression, effective spatial and border perception is more essential for locating objects of varying sizes and orientations. As depicted in Fig. 1(b), inaccurate angle prediction can lead to poor detection results. Therefore, it is essential to design task-specific structures that decouples classification and regression regarding their different feature requirements.

In this article, we propose a task-specific heterogeneous (TSH) network for object detection in aerial images. Our method effectively decouples the classification and localization tasks, taking into account both the characteristics of aerial images and task-specific attributes. As shown in Fig. 2, to reduce the background and interclass interference, we design an interference-suppression module (ISM). In the

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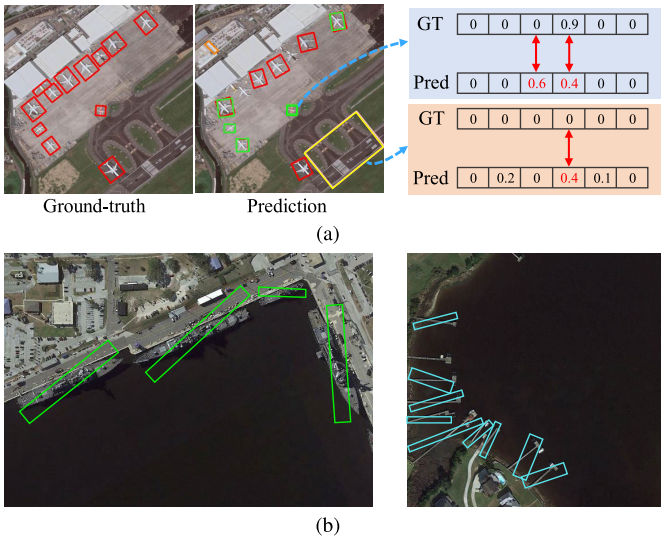


Fig. 1. (a) Inconsistency between ground-truth labels and predicted classification confidence. We use the joint representation in GFL as the ground-truth label. The yellow box is the false positive caused by background interference. The green boxes denote the misclassified objects. (b) Poor detection results of arbitrary-oriented objects due to inaccurate angle prediction.

ISM, we first enhance the responses of objects while suppressing the background interference in spatial domain. Then, an interclass channel decomposing (CD) module is proposed to separate the features of different categories in channel dimension, ensuring that the feature channels are class-aligned. In addition, we employ an interference-constraint (IC) loss to explicitly supervise the feature representation learning, leading to more discriminative features for classification. After that, we devise a joint-learning quality estimation (JQE) module that adaptively weighs the contributions of classification and localization features, boosting the estimation of localization quality. For regression, we design a point-based localization (PBL) branch, leveraging the advantages of point set representation to handle aerial objects with diverse shapes and orientations. To promote the localization accuracy, we utilize a dynamic information aggregation (DIA) module to model the relationships between the dispersible points, resulting in more accurate position of contour points.

In summary, our main contributions can be described as follows.

1) A TSH network is developed considering both the characteristics of aerial images and specific attributes of classification and localization tasks.
2) An ISM is proposed to reduce both the background and interclass interference, followed by a JQE module to estimate reliable localization quality.
3) A PBL branch is designed to produce accurate location of aerial objects based on point set representation.

II. Related Work

A. General Object Detection

Object detectors based on convolutional neural network (CNN) typically fall into two main categories: one-stage detectors and two-stage detectors. Two-stage detectors follow a two-step process, where they initially generate a set of candidate proposals and subsequently perform classification and regression on these region features to obtain the final detection results. For example, Faster regions with CNN features (R-CNN) [17] is a classic two-stage framework that incorporates a region proposal network (RPN) to facilitate the generation of region proposals. Dynamic R-CNN [18] adapts the label assignment criteria and regression loss function to make better use of the training samples. Decoupled R-CNN [19] resorts to a decoupled sampling strategy to select two different sets of proposals from RPN for classification and regression tasks, respectively. Deformable template network (DTN) [20] generates a deformable template for an object and then performs part alignment to handle object deformation. disentangled R-CNN (DR-CNN) [21] extends the idea of disentangled dense object detector (DDOD) [22] to two-stage detectors, disentangling the detector into three aspects, i.e., input feature, label assignment, and pyramid supervision. All of these two-stage detectors rely on candidate proposals, which often results in increased computational time. In contrast, one-stage detectors perform classification and regression directly on the dense feature maps without region proposals. For example, RetinaNet [23] introduces a novel focal loss (FL) to handle the imbalanced positive and negative samples, boosting the training of one-stage detectors. Fully convolutional one-stage object detector (FCOS) [24] adopts a simple anchor-free design, treating object detection as a pixelwise prediction task and eliminating the need for complex anchor-related computations. ObjectBox [25] tackles the label assignment issue for objects with varying sizes and shapes, resulting in a more robust and generalizable detector across different datasets. YOLOv1 [26] is the first one-stage detector with an efficient architecture and remarkably fast inference speed. Subsequent versions of you only look once (YOLO) [27], [28], [29], [30] further promote the detection accuracy. Notably, YOLOv7 [31] focuses on optimizing specific modules and training strategies, leading to state-of-the-art performance in both inference speed and detection accuracy.

Recently, researchers have identified the task misalignment between classification and localization and have been working on decoupling these two tasks. For example, double-head R-CNN [32] and task-aware spatial disentanglement (TSD) [33] revisit the widely used sibling head in R-CNN family. In one-stage detectors, the decoupled detection head with two parallel branches is commonly used for extracting classification and localization features separately. Furthermore, DDOD [22] proposes to adaptively learn specific features for each task via separate deformable convolutional offset for each branch. YOLOX [34] also analyzes the effect of coupled detection head and introduces a decoupled head to the YOLO family for the first time.

Despite the attempt to decouple classification and regression tasks, they are not entirely independent in the detection framework, as the classification score is often used as a measure of localization accuracy during inference. Therefore, some methods aim to improve the consistency between these two factors. For example, intersection-of-union (IoU) balance [35] and prime sample attention (PISA) [36] employ the localization quality of samples to determine the weights in classification loss, enhancing the correlation between classification and localization. In contrast, IoU-Net [37] and FCOS [24] estimate the localization quality in the form of IoU or centerness score

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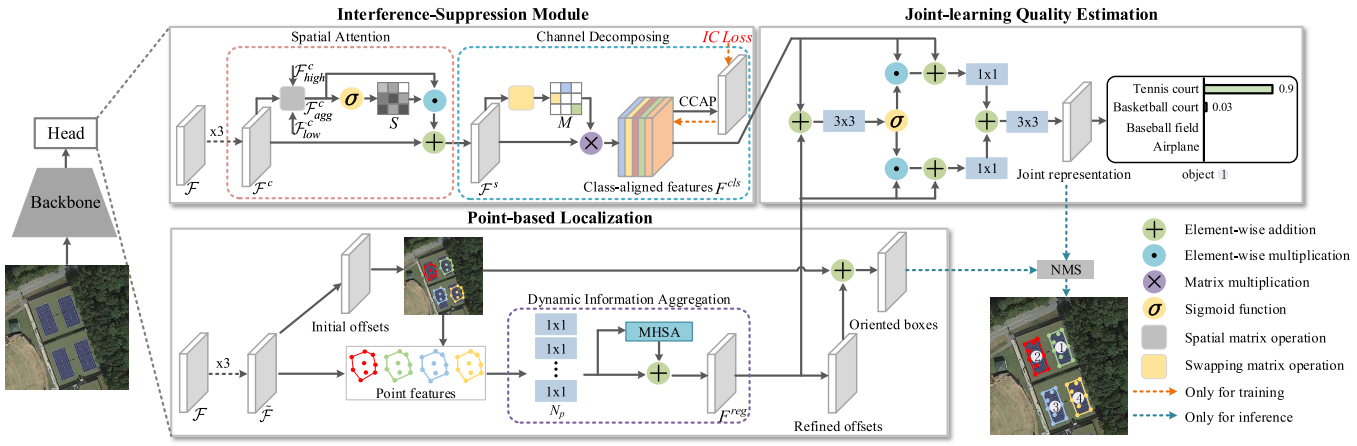


Fig. 2. Overall architecture of the proposed TSH. An input image is first fed to the backbone network for feature extraction and then passed to detection head with two parallel branches. The ISM consists of an SA module and a CD module, reducing the background and interclass interference, respectively. CCAP represents cross-channel average pooling. The IC loss imposed directly on the feature maps after CCAP provides supervision for explicitly reducing the interference. Then, the JQE module adaptively combines the classification and regression features to predict more reliable confidence. As for object localization, the PBL branch employs adaptive points and DIA to produce accurate oriented boxes. MHSA denotes multihead self-attention. NMS denotes nonmaximum suppression postprocessing during inference.

via a separate branch, which is then combined with the classification confidence during inference. As an alternative approach, generalized focal loss (GFL) [38] combines the localization quality, and classification score into a joint representation and proposes a quality FL (QFL) for optimizing this representation.

B. Object Detection in Aerial Images

In aerial images, objects often exhibit arbitrary shapes and orientations, resulting in severe misalignment for horizontal object detection methods. Therefore, researchers have developed lots of methods to predict the OBB. For instance, rotation region proposal networks (RRPN) [39] adopts rotated anchors with different angles, scales, and aspect ratios to better cover the oriented objects. Region of interest (RoI) Transformer [2] proposes to transform horizontal RoIs into rotated RoIs. Later, oriented R-CNN [40] replaces the RPN with a lightweight oriented RPN, significantly improving the efficiency of oriented proposal generation. On this basis, dual-aligned oriented detector (DODet) [41] further proposes a localization-guided detection head to obtain aligned features for more accurate classification. Besides, some methods try to address the feature misalignment and extract rotation-invariant features. For instance, dynamic refinement network (DRN) [42] proposes a rotation convolutional layer that can adaptively adjust the receptive fields. Refined rotation retinaNet (R^3Det) [4] reconstructs the feature map by enhancing the feature points with information of corresponding rotated proposals.

Some methods utilize various OBB representations to better describe the oriented objects, along with more effective loss functions to facilitate network optimization. For instance, detector for small, cluttered and rotated objects (SCRDet) [43] alleviates the boundary issue of oriented box regression via an improved smooth $L1$ loss. Gliding vertex [5] predicts gliding distances between vertex of oriented boxes and horizontal proposals. Jiang et al. [44] decode the orientation information by orthogonally decomposing the long-edge vector. Box boundary-aware vectors (BBAVectors) [45] learns additional width and height of external horizontal boxes instead of directly regressing the angle parameter. Queuing people detector (QPDet) [46] draws on the idea of polar coordinates and employs the radius and two rotation offsets to locate the vertices of oriented boxes. By using a CSL [12] and Densely Coded Label (DCL) [47] predict the angle through classification instead of direct regression, avoiding the periodicity issue of angle. Phase-shifting coder (PSC) [48] analyzes the common issues in angle regression and proposes a PSC to estimate more accurate orientation. In addition, some researchers focus on designing specific label assignment strategies for oriented object detection. For example, dynamic anchor learning (DAL) [49] proposes a new matching degree instead of IoU to evaluate the quality of anchors and assigns training samples according to the matching degree, achieving better performance. GGHL [15] assigns the samples based on the oriented Gaussian heat map and dynamically adjusts the soft weight of samples according to the object's shape. Dynamic coarse-to-fine learning (DCFL) [50] explores oriented tiny object detection and leverages the dynamic prior to assign more high-quality positive samples. Doloriel and Cajote [51] design an attention-points network that integrates two loss functions to improve the learning of small oriented objects. Zhou et al. [52] reduce information loss in upsampling and makes better use of local contextual features to detect small objects.

However, these methods utilize regression-based approach for oriented box prediction and lack effective spatial and boundary perception, which may lead to inaccurate localization. Therefore, some works focus on key-point-based detection methods to improve the localization accuracy. For example, center-head point extraction based detector (CHPDet) [53] designs a network for ship detection that utilizes center point and size information to locate ships and regresses the head points to determine their orientations. Oriented objects detection network (O^2-DNet) [54] estimates the locations of two vertical middle lines and their intersection point for corresponding OBB, and each line corresponds to two endpoints. Faster R-CNN-estimation (FR-est) [55] uses a set of uniformly distributed points within the oriented boxes to represent objects and improves the

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accuracy of point localization through coarse-to-fine learning. Orientation-sensitive keypoint based rotated Detector (OSKDet) [56] utilizes a set of key points to represent the rotated box and proposes an orientation-sensitive heat map to learn the locations and directions of key points. Oriented RepPoints [57] generates the point set by predicting offsets toward the oriented object and designs a quality assessment and assignment scheme to facilitate the learning of points.

Different from these methods, the proposed TSH employs a heterogeneous structure that considers the different requirements and characteristics of classification and regression tasks in aerial images, respectively. Thus, the detector can learn more effective task-specific features to promote the detection accuracy.

C. Feature Representation Learning

Effective feature representation is crucial for distinguishing foreground objects from the background. However, aerial images often contain complex backgrounds that may introduce a large amount of noise, impairing the representation ability of features. To remedy this issue, some works adopt feature enhancement and noise reduction to emphasize features of important regions while suppressing irrelevant ones. For example, Zhao et al. [58] use a gated network for salient object detection, allowing to adaptively control the flow of interference and useful contextual information. Attention guided contextual feature fusion network (ACFFNet) [59] integrates a channel attention module into the cross-level feature fusion process to enhance relevant information for salient objects. Deng et al. [60] propose a foreground attention reweighting method to prioritize foreground instances and suppress potential noise in the background region. Foreground refinement network (FoRDet) [61] enhances the discrimination of foreground regions by utilizing foreground modeling to aggregate relevant contextual information. iffDetector [62] leverages high-level semantic information to filter the background noise in feature level during training and inference. Shen et al. [63] enhance the feature pyramid in object detection by integrating global semantic and critical location information of hierarchical feature maps. Critical feature capturing network (CFCNet) [64] utilizes a polarization attention module to generate more discriminative features for classification. Xie et al. [65] propose a feature denoising block to enhance adversarial robustness. Inspired by this, SCRDet++ [66] further develops an instance-level denoising module to reduce the interference and improves the detection accuracy of small and cluttered objects.

These methods typically emphasize object features through adaptive feature fusion or attention mechanisms, treating it only as a binary classification task. Although SCRDet++ adopts an attention map to decouple features of different object categories into separate channels, it lacks explicit supervision on the enhanced feature maps. In this article, we propose a supervised way to effectively suppress both background and interclass interference, including an IC loss directly imposed on the class-aligned features to ensure more reliable and accurate results.

III. Methodology

The overall architecture of the proposed TSH is illustrated in Fig. 2. It is composed of four parts as follows: 1) the backbone network; 2) the ISM; 3) the JQE module; and 4) the PBL branch. First, the backbone network ResNet50-feature pyramid network (FPN) (R-50-FPN) [67], [68] intakes the input image to generate hierarchical features. Then, the features are fed into the ISM to reduce the interference. After that, the quality estimation module integrates the classification feature and regression feature to estimate the localization quality. Moreover, the PBL branch conducted on each hierarchical feature is designed to produce accurate detection of arbitrary-oriented objects. Sections III-A–III-C provides a detailed description of the proposed components.

A. Interference-Suppression Module

Aerial images usually capture a large area of land containing various and complex spatial distributions of ground objects in the scenes. This may lead to a large amount of noise for the classification task at pixel level. In addition, for multicategory object detection, the detector requires strong discrimination capability to address interclass feature coupling of objects with similar appearances. Therefore, we consider it essential to reduce both the background and interclass interference. To this end, we design an ISM to improve the feature representation capability for classification. Specifically, we first suppress the background interference while enhancing the responses of objects in spatial domain. Then, we perform interclass CD by splitting the features into groups from channel dimension, corresponding to different object categories. Finally, an IC loss is proposed to explicitly enforce the discriminability of the decomposed features.

The detailed structure of ISM is illustrated in Fig. 2. Following [24], we utilize the pyramid features $P3$–$P7$ extracted from the backbone R-50-FPN. Then, for each feature map $\mathcal{F} \in \mathbb{R}^{C\times H\times W}$ of the pyramid, where H, W, and C are the height, width, and number of channels, respectively, we first stack three consecutive 3×3 convolutions, each with C filters, and followed by rectified linear unit (ReLU) activation. The obtained feature is denoted as $\mathcal{F}^c \in \mathbb{R}^{C\times H\times W}$. C is set to 256 by default. Then, we exploit a spatial attention (SA) module to reweight the feature $\mathcal{F}^c$, highlighting the responses of objects while suppressing the uninformative background. Motivated by Li et al. [69], we generate the saliency map based on the fused features, followed by a convolution operation with a sigmoid function, as shown in Fig. 3(a). Concretely, we aggregate the features across levels to focus on more discriminative regions. The two features $\mathcal{F}^c_{\text{low}} \in \mathbb{R}^{C\times 2H\times 2W}$ and $\mathcal{F}^c_{\text{high}} \in \mathbb{R}^{C\times H/2\times W/2}$ from adjacent level of $\mathcal{F}^c$ are resized to the same resolution as $\mathcal{F}^c$ through bilinear interpolation. Then, the three features are fed to a 3×3 convolution and are aggregated via an addition operation, followed by a 1×1 convolution to obtain the feature $\mathcal{F}^c_{\text{agg}}$. A sigmoid function $\sigma(\cdot)$ is used to generate the saliency map $S = \sigma(\mathcal{F}^c_{\text{agg}})$. Thus, the feature of salient parts containing foreground objects is enhanced adaptively via a multiple operation. Besides, a residual connection is adopted to preserve the original information.

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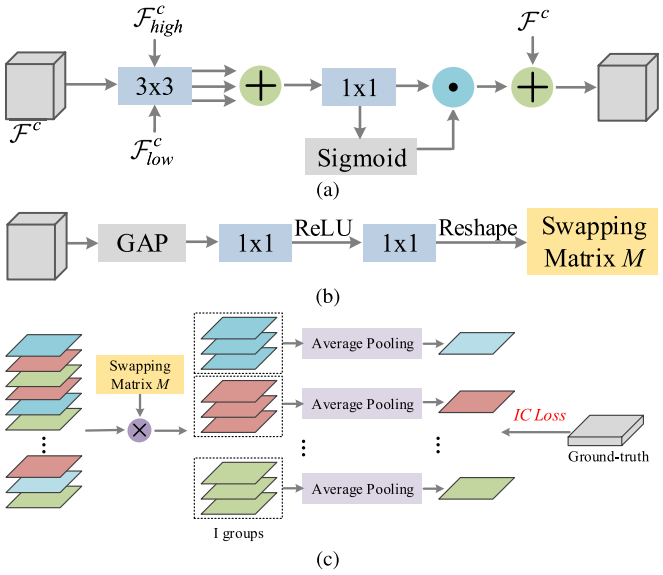


Fig. 3. (a) Structure of SA module. (b) Structure to generate the swapping matrix. (c) Process of decomposing feature channels and adding the IC loss.

Therefore, the enhanced feature $\mathcal{F}^s$ can be obtained as follows:

$$\mathcal{F}^s = S \times \mathcal{F}^c_{\text{agg}} + \mathcal{F}^c. \tag{1}$$

Next, we conduct the interclass CD on $\mathcal{F}^s$. To group the feature channels corresponding to each category together, we perform channel swapping on $\mathcal{F}^s$ as follows:

$$\mathcal{F}' = R^{chw}\left(M \otimes R^{c(hw)}\left(\mathcal{F}^s\right)\right) \tag{2}$$

where $\otimes$ denotes matrix multiplication. $R^{c(hw)}$ reshapes the input into the size of $C \times HW$, and R^{chw} reshapes it into the size of $C \times H \times W$. M is a learnable matrix of size $C \times C$ generated as follows:

$$M = \mathcal{H}\left(\mathcal{F}^s\right) \tag{3}$$

whose goal is to achieve channel swapping. The implementation of $\mathcal{H}$ is illustrated in Fig. 3(b). We expect the homogeneous feature maps to be compact along channel dimension after channel swapping. Furthermore, $\mathcal{F}'$ is divided into I groups from channel dimension, where I indicates the number of classes. Each class is represented by ξ feature channels, and $C = I \times \xi$. The feature channels corresponding to the ith class are represented by $\mathbf{F}^{\text{cls}}_i \in \mathbb{R}^{\xi \times HW}$. We assume that $\mathbf{F}^{\text{cls}}_i$ only carries the effective semantic information of the objects assigned to the corresponding class i. Then, $\mathbf{F}^{\text{cls}} = \{\mathbf{F}^{\text{cls}}_1, \mathbf{F}^{\text{cls}}_2, \ldots, \mathbf{F}^{\text{cls}}_I\}$ is passed to the subsequent localization quality estimation module.

Moreover, to ensure that the feature channels are class-aligned and suppress the noise, we propose the IC loss to supervise the learning of $\mathbf{F}^{\text{cls}}$. As shown in Fig. 3(c), the average response of feature channel in $\mathbf{F}^{\text{cls}}_i$ is computed by a cross-channel average pooling (CCAP); thus, we can obtain $\mathbf{F}' \in \mathbb{R}^{I \times HW}$, where each vector of size HW corresponds to a particular class. Then, the IC loss is computed as follows:

$$\mathcal{L}_{\text{IC}} = \mathcal{L}(\mathbf{Y}, \sigma(\mathbf{F}')) \tag{4}$$

where $\mathbf{Y}$ denotes the ground-truth class label, and we adopt FL as the function $\mathcal{L}(\cdot, \cdot)$ in this work. As we introduce CCAP to acquire the average response of feature channels belonging to each class, the IC loss imposed directly on the feature map can suppress the features of background and other classes. Thus, the enhanced representation $\mathbf{F}^{\text{cls}}$ is class-aligned and more discriminative for classification.

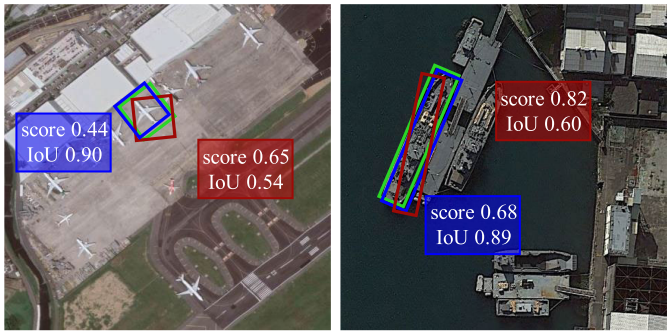


Fig. 4. Illustration of the consistency between classification score and localization quality. Blue boxes with higher IoU have lower classification scores.

B. Joint-Learning Quality Estimation

Apart from separating foreground proposals from the background and classifying them into the right category, we also need to predict the localization accuracy of the detection box. As shown in Fig. 4, the classification scores cannot accurately reflect the localization quality. Although some methods predict additional IoU [37] or centerness [24] scores as the localization quality, they are learned separately from classification scores, which may degrade the detection performance. To address this issue, GFL [38] predicts a joint representation of localization quality and classification, supervised by a continuous version of FL. However, due to the lack of localization information, the learned joint representation based only on the classification feature is not accurate enough. Therefore, we exploit a JQE module to improve the prediction of joint representation, which integrates the regression feature with enhanced classification feature.

Fig. 2 illustrates the structure of this module. In practice, for the feature $\mathbf{F}^{\text{cls}}$ extracted from the classification branch and the feature $\mathbf{F}^{\text{reg}}$ produced by the regression branch, we first fuse the two features via an addition operation. Then, the fused feature is passed to a 3×3 convolution with a sigmoid function to generate the attention A, which can be given by

$$A = \sigma\left(\text{conv}\left(\mathbf{F}^{\text{cls}} + \mathbf{F}^{\text{reg}}\right)\right). \tag{5}$$

The attention A is used to enhance the classification and regression features by capturing their correlation. In addition, to preserve the original information, the enhanced features are combined with their original features via a residual connection. Then, a 1×1 convolution is used to transform the features, which can be formulated as follows:

$$\begin{aligned} \hat{\mathbf{F}}^{\text{cls}} &= \text{conv}_1(\mathbf{F}^{\text{cls}} + A \cdot \mathbf{F}^{\text{cls}}) \\ \hat{\mathbf{F}}^{\text{reg}} &= \text{conv}_1(\mathbf{F}^{\text{reg}} + A \cdot \mathbf{F}^{\text{reg}}). \end{aligned} \tag{6}$$

The two features are further aggregated via an addition operation. Then, the aggregated feature will go through a 3×3 convolution to estimate the localization quality $\widehat{\mathbf{Q}} \in \mathbb{R}^{C \times H \times W}$. The learning of this joint representation is supervised by a QFL [38]. More importantly, our

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module can adaptively integrate the classification and localization information to facilitate the estimation of localization quality.

C. Point-Based Localization

Objects in aerial images usually appear in arbitrary orientations. To localize these objects, some methods generate the oriented boxes by predicting an extra angle parameter based on vanilla detectors. However, inaccurate orientation prediction may lead to a significant change in IoU, especially for objects with large aspect ratios, as shown in Fig. 5(a). To alleviate this issue, we design a PBL branch. In the PBL, we adopt adaptive point set representation to handle aerial objects with diverse shapes and orientations. Moreover, to obtain more accurate locations of key points, we employ a DIA module to model the relationships between these dispersible points, thus enriching their feature representations. Based on the correlative dependence, each point adaptively gathers information from associated locations and then evolves toward the contour point more precisely. Fig. 5(b) shows the evolution of these adaptive points.

The structure of the PBL is illustrated in Fig. 2. It consists of two stages that progressively refine the adaptive points. The initial stage generates the point sets by refining from the object center point on each feature map. The second stage predicts the offsets of new sampled points with respect to the old ones. Specifically, given feature map $\mathcal{F}$ from the backbone, the branch first applies three consecutive 3×3 convolutions, each with C filters, followed by ReLU activation to obtain feature $\widetilde{\mathcal{F}} \in \mathbb{R}^{C\times H\times W}$. Then, a 3×3 convolution with $2N_p$ filters is performed to compute offsets for a set of adaptive sampled points. N_p is the number of sampled points. Thus, the position $\mathbf{P} \in \mathbb{R}^{N_s\times N_p\times 2}$ of N_p points for $N_s = H \times W$ spatial locations in feature $\widetilde{\mathcal{F}}$ can be obtained according to the $2N_p$ offset field (i.e., the x offset and y offset).

Then, we extract the initial control point feature $\mathbf{F}^{\text{init}} \in \mathbb{R}^{N_s\times N_p\times C}$ from corresponding location in $\widetilde{\mathcal{F}}_l$ by bilinear interpolation. As shown in Fig. 5(c), we conduct N_p independent 1×1 convolution on $\mathbf{F}^{\text{init}}$ to project the initial feature into deeper space, which can be formulated as follows:

$$\mathbf{F}_{ij}^{\text{conv}} = W_j \otimes \mathbf{F}_{ij}^{\text{init}} \tag{7}$$

where $\mathbf{F}_{ij}^{\text{init}} \in \mathbb{R}^C$ and $\mathbf{F}_{ij}^{\text{conv}} \in \mathbb{R}^C$ denote the jth point feature corresponding to the ith spatial location of $\mathbf{F}^{\text{init}}$ and the output feature $\mathbf{F}^{\text{conv}} \in \mathbb{R}^{N_s\times N_p\times C}$, respectively. $W_j \in \mathbb{R}^{C\times C}$, $j \in 1, \ldots, N_p$, represents the kernel weights with regard to the index of sampled point j. After that, we adopt multihead self-attention (MHSA) to mine the relationship between N_p associated points and propagate the information among the same instance. The aggregated feature representation $\mathbf{F}^{\text{att}}$, thus, can be expressed as follows:

$$\mathbf{F}^{\text{att}} = \text{MultiHead}(\mathbf{F}^{\text{conv}}). \tag{8}$$

Also, the attention weights are computed as in [70]

$$\text{Attn}(\mathbf{F}^{\text{conv}}) = \text{softmax}\left(\frac{Q(\widehat{\mathbf{F}})K(\widehat{\mathbf{F}})^T}{\sqrt{C}}\right)V(\mathbf{F}^{\text{conv}}) \tag{9}$$

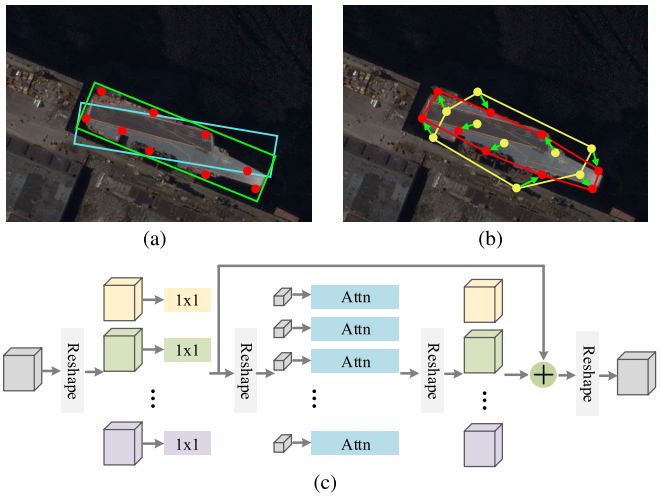


Fig. 5. (a) Illustration of different OBB representations. The light blue box represents inaccurate detection produced by angle-based regression method. The green box denotes the converted oriented box based on the learned adaptive points. (b) Evolution process of learned points after applying the DIA. (c) Structure of the DIA module. Attn represents MHSA.

where Q, K, and V denote different linear layers. Note that the values do not contain positional information, while keys and queries are the same denoted as $\widehat{\mathbf{F}}$, which integrates position embedding into $\mathbf{F}^{\text{conv}}$ to reserve location information of the sampled features. $\widehat{\mathbf{F}}$ is given by

$$\widehat{\mathbf{F}}_{ij} = \varphi\left(\mathbf{P}_{ij}\right) + \mathbf{F}_{ij}^{\text{conv}} \tag{10}$$

where φ indicates the sine positional encoding function and $\mathbf{P}_{ij}$ represents the coordinate of sampled point.

Moreover, we add the aggregated feature $\mathbf{F}^{\text{att}}$ into the original feature $\mathbf{F}^{\text{conv}}$ to preserve the original feature information

$$\mathbf{F}^{\text{fused}} = \mathbf{F}^{\text{conv}} + \alpha\mathbf{F}^{\text{att}} \tag{11}$$

where α denotes the parameter and is set as 1 by default. Then, the outputs of N_p points at each spatial location are combined to generate final regression feature

$$\mathbf{F}_i^{\text{reg}} = \sum_{j=1}^{N_p} \mathbf{F}_{ij}^{\text{fused}}. \tag{12}$$

We further reshape the refined key-point feature $\mathbf{F}^{\text{reg}}$ into the size of $C \times H \times W$ and then predict the key-point offset $\mathbf{O} \in \mathbb{R}^{2Np\times H\times W}$ via the regressor $\mathcal{R}$

$$\mathbf{O} = \mathcal{R}(\mathbf{F}^{\text{reg}}) \tag{13}$$

where $\mathcal{R}$ is implemented by a 1×1 convolution. Finally, the key-point location $\widehat{\mathbf{P}} \in \mathbb{R}^{N_s\times N_p\times 2}$ is computed by

$$\widehat{\mathbf{P}} = \mathbf{P} + \mathbf{O} \tag{14}$$

where we reshape the size of $\mathbf{O}$ to be consistent with $\mathbf{P}$. We use $N_p = 9$ by default in our experiments. The point set representation is capable of adaptively capturing the geometric structure of objects with varying shapes and orientations. Furthermore, through modeling the relationships between the dispersible points, our model can produce more accurate position of contour points.

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D. Training and Inference

During training, we follow the same label assignment mechanism in rotated object detector based on FCOS (FCOS-R) [71] to assign the ground-truth labels. Then, the whole training objective function for our method is given by

$$\mathcal{L} = \mathcal{L}_{\mathrm{QE}} + \mathcal{L}_{\mathrm{REG}} + \lambda \mathcal{L}_{\mathrm{IC}} \tag{15}$$

where λ is the weight factor of IC loss. $\mathcal{L}_{\mathrm{QE}}$ and $\mathcal{L}_{\mathrm{REG}}$ represent the loss for localization quality estimation and regression, respectively. We adopt QFL to supervise the learning of JQE module. Also, the formulation of $\mathcal{L}_{\mathrm{REG}}$ is consistent with [57]. In the inference stage, MinAeraRect is employed to convert the learned point set into OBBs of objects.

IV. Experiments

A. Experimental Setting

1) Datasets: To comprehensively evaluate the performance of our TSH, we perform experiments on several public aerial object detection datasets, which cover a range of scenes and diverse objects with different shapes and categories.

1) *DIOR-R [72], [73]:* This dataset is a multiclass aerial object detection dataset containing 190 288 objects of 20 categories. This dataset contains 23 463 images with a resolution of 800 × 800 pixels, of which 5862 images are used for training, 5863 images for validation, and 11 738 images for testing. In our experiments, the models are trained on the training and validation sets and evaluated on the test set.
2) *DOTA-1.0 [74]:* This dataset is widely used for evaluating the performance of aerial object detection methods. It contains 2806 images ranging from 800 × 800 pixels to 4000 × 4000 pixels, with more than 188 000 objects covering 15 categories. The dataset is split into 1411 training images, 458 validation images, and 937 testing images. For training, we use the training and validation sets, and the test set is used for evaluation. Due to the large size of the images, they are cropped into patches of 1024 × 1024 pixels with an overlap of 200 pixels.
3) *HRSC2016 [75]:* This dataset consists of 436 training images, 181 validation images, and 444 testing images, ranging in size from 300 × 300 pixels to 1500 × 900 pixels.
4) *UCAS-AOD [76]:* It contains 1510 aerial images of approximately 659 × 1280 pixels, including two categories of 14 596 instances. In line with [64], we randomly split the dataset into training, validation, and test sets with a ratio of 5:2:3.

2) Evaluation Metric: We use mean average precision (mAP) as the evaluation metric for our model, which is the average of area under the precision–recall curve (AP) over all classes, with the IoU threshold set to 0.5. To speed up AP calculation, two approximate methods are employed, one is the 11-point interpolation method (PASCAL VOC07 [77] evaluation metric), and the other is full interpolation method (PASCAL VOC12 [78] evaluation metric). Unless otherwise specified, we report mAP under the VOC07 evaluation metric.

TABLE I
Ablation Studies of Different Components on the DIOR-R Dataset

Baseline (FCOS-R)	ISM	JQE	PBL	mAP(%)
✓				63.37
✓	✓			64.28
✓		✓		64.12
✓			✓	65.02
✓	✓	✓		64.75
✓	✓		✓	65.53
✓		✓	✓	65.60
✓	✓	✓	✓	**66.17**

3) Implementation Details: We conduct the experiments on Sugon X785-G30 server with Intel Xeon Gold 6230 CPU and 256G RAM. The operating system is CentOS 7.7. Our models are built with PyTorch [79] based on the MMDetection [80] and MMRotate libraries [81]. We adopt R-50-FPN [67], [68] as the backbone network in all our experiments. Stochastic gradient descent (SGD) is employed to optimize the models. The batch size is 8 with two images per GPU. The initial learning rate is set to 0.005 and is divided by 10 at each decay step. The weight decay and momentum are set as 0.0001 and 0.9, respectively. We train the models on rotated object detection in optical remote sensing images (DIOR-R), dataset for object detection in aerial images version 1.0 (DOTA-1.0), high resolution ship collection 2016 (HRSC2016), and University of Chinese Academy of Sciences-aerial object detection (UCAS-AOD) datasets for 36, 36, 72, and 72 epochs, respectively. The learning rate decays at the 24th and 33rd epochs for DIOR-R and DOTA-1.0 datasets and the 48th and 66th epochs for HRSC2016 and UCAS-AOD, datasets, respectively. Random flipping and random rotation are employed for data augmentation.

B. Ablation Study

To verify the effectiveness of the proposed TSH, we conduct a comprehensive ablation study on the DIOR-R dataset. FCOS-R [71] is employed as the baseline method that extends FCOS [24] to generate the OBBs by predicting an extra angle parameter. Specifically, FCOS-R utilizes R-50-FPN [67], [68] as backbone to extract multiscale features from pyramid levels $P3$–$P7$; then, the features at each level are fed into the detection head with two parallel branches for classification and regression, respectively. In the proposed method, we introduce ISM and JQE to replace the original classification branch. ISM replaces the four convolutional layers for interference suppression, and JQE displaces the final classification layer to predict a joint representation. Noting that JQE combines classification and regression features to enhance the learning of the joint representation. In addition, we use PBL instead of the original regression branch, which learns adaptive points to improve the localization accuracy of oriented objects. In ablation study, we add the key components to the baseline progressively while keeping other conditions unchanged.

1) Evaluation of Different Modules: As shown in Table I, the baseline FCOS-R achieves the mAP of 63.37%. When

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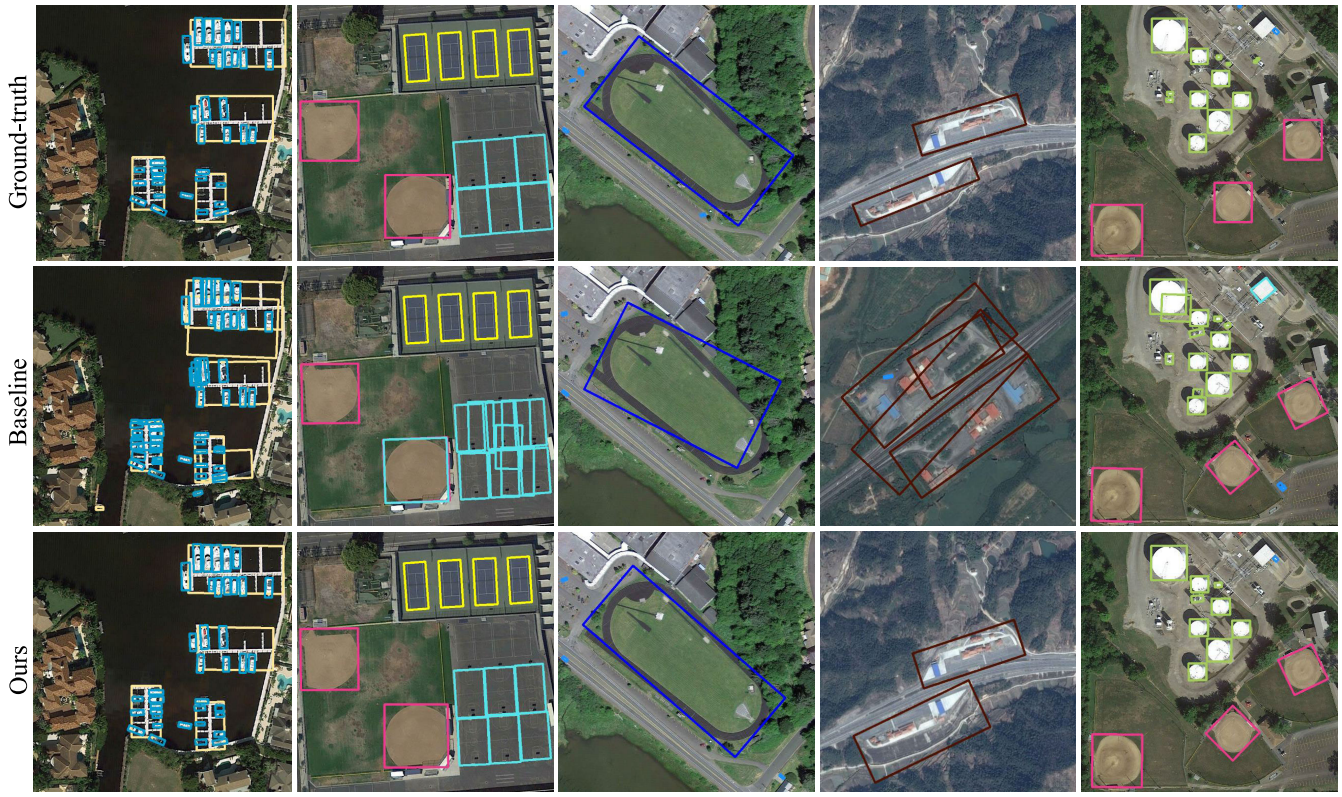


Fig. 6. Comparison of our method with the baseline FCOS-R [71] on the DIOR-R dataset. The boxes with different colors represent objects of different categories.

TABLE II

ABLATION STUDIES OF SUBMODULES IN ISM, JQE, AND PBL ON THE DIOR-R DATASET. SA, CD, AND $\mathcal{L}_{IC}$ DENOTE THE SPATIAL ATTENTION, CHANNEL DECOMPOSING, AND IC LOSS IN ISM, RESPECTIVELY. FL, QFL, FI, AND AFM STAND FOR THE FOCAL LOSS, QUALITY FOCAL LOSS, FEATURE INTERACTION, AND ADAPTIVE FUSION MODULE IN JQE, RESPECTIVELY. PR, DEFORMCONV, AND DIA REPRESENT THE POINT REPRESENTATION, DEFORMABLE CONVOLUTION, AND DYNAMIC INFORMATION AGGREGATION IN PBL, RESPECTIVELY

	Baseline (FCOS-R)	SA	CD	$\mathcal{L}_{IC}$	mAP(%)
ISM	✓				63.37
	✓	✓			63.72
	✓		✓		63.38
	✓		✓	✓	63.95
	✓	✓	✓	✓	64.28
	FL	QFL	FI	AFM	mAP(%)
JQE	✓				62.57
		✓			63.37
		✓	✓		63.75
		✓		✓	64.12
	Baseline (FCOS-R)	PR	DeformConv	DIA	mAP(%)
PBL	✓				63.37
	✓	✓	✓		64.19
	✓	✓		✓	65.02

TABLE III

COMPARISON RESULTS OF ISM WITH THE EXISTING FEATURE ENHANCEMENT STRATEGIES ON THE DIOR-R DATASET

Methods	Feature Enhancement	mAP(%)
Baseline (FCOS-R)	None	63.37
	iffDetector [62]	60.49
	Non-Local [65]	62.38
	InLD [66]	63.57
	CFC-Net [64]	63.71
	Bilateral Filter [65]	63.79
	Ours	64.28

TABLE IV

EVALUATION OF USING VARYING λ IN ISM. λ DENOTES THE WEIGHT FACTOR OF IC LOSS, WHICH CONTROLS THE STRENGTH OF THE IC. LARGER λ LEADS TO STRONGER INTERFERENCE SUPPRESSION

λ	0.0	0.1	0.3	0.5	0.7	0.9	1.0
mAP(%)	63.38	63.56	63.78	**63.95**	63.79	63.72	63.65

we add the ISM, the performance is improved from 63.37% to 64.28%. It indicates that the proposed ISM is effective to suppress interference and promote the detection accuracy. We then apply the PBL, and the results show that the PBL branch improves the mAP by 1.65% compared with the baseline. It demonstrates that adaptive point learning is beneficial to detect aerial objects with arbitrary shapes and orientations. This is because the adaptive points are more flexible to capture the geometric structure and semantic information of aerial objects. Correspondingly, performing task decoupling with ISM and PBL modules can achieve 2.16% mAP improvement, demonstrating the effectiveness of the

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TABLE V

COMPARISON RESULTS WITH RELATED METHODS ON THE DIOR-R DATASET. EACH COLUMN'S BEST AND SECOND-BEST RESULTS ARE INDICATED BY RED AND BLUE COLORS, RESPECTIVELY. R-50-FPN REPRESENTS RESNET50-FPN. R-101-FPN REPRESENTS RESNET101-FPN. CONVNEXT-T-FPN DENOTES CONVNEXT-TINY [82] WITH FPN. SWIN-T-FPN DENOTES SWIN TRANSFORMER-TINY [83] WITH FPN. THE RESULTS OF RELATED METHODS ARE COLLECTED FROM ORIGINAL PAPERS, AND "†" DENOTES OUR REPRODUCED RESULT

Methods	Backbone	Image Size	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP(%)	FPS
Two-stage Methods																								
CSL† [12]	R-50-FPN	800×800	62.29	26.90	71.17	81.35	26.28	72.69	73.22	57.15	20.14	75.97	76.26	34.53	40.05	75.28	68.24	54.50	81.35	48.26	38.47	63.60	57.38	15.9
Faster RCNN-O [17]	R-50-FPN	800×800	62.79	26.80	71.72	80.91	34.20	72.57	18.95	66.45	65.75	66.63	79.24	34.95	34.95	34.95	64.34	71.21	81.44	47.31	50.46	65.21	59.54	16.2
Gliding Vertex [5]	R-50-FPN	800×800	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.70	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06	15.4
RoI-Transformer [2]	R-50-FPN	800×800	63.34	37.88	71.78	87.53	40.68	72.60	26.86	78.71	68.09	68.96	82.74	47.71	55.61	81.21	78.23	70.26	81.61	54.86	43.27	65.52	63.87	13.9
QPDet [46]	R-50-FPN	800×800	63.22	41.39	71.97	88.55	41.23	72.63	28.82	78.90	69.00	70.07	83.01	47.83	55.54	81.23	72.15	62.66	89.05	58.09	43.38	65.36	64.20	15.3
AOPG [73]	R-50-FPN	800×800	62.39	37.79	71.62	87.63	40.90	72.47	31.08	65.42	77.99	73.20	81.94	42.32	54.45	81.17	72.69	71.31	81.49	60.04	52.38	69.99	64.41	13.8
Oriented R-CNN [40]	R-50-FPN	800×800	62.00	44.92	71.78	87.93	43.84	72.64	35.46	66.39	81.35	74.10	80.95	43.52	58.42	81.25	68.01	62.52	88.62	59.31	43.27	66.31	64.63	15.5
ReDet† [84]	R-50-FPN	800×800	63.31	43.30	72.02	89.08	43.44	80.12	80.06	70.35	28.58	70.10	83.07	48.02	57.57	81.25	67.58	62.65	81.49	56.70	48.48	66.24	64.67	10.7
CGCDet [85]	R-50-FPN	800×800	68.46	38.34	79.12	86.21	38.97	73.52	26.84	66.00	74.72	67.49	84.45	48.02	56.05	81.25	79.32	72.17	88.56	50.69	51.72	65.84	64.88	-
DODet [41]	R-50-FPN	800×800	63.40	43.35	72.11	81.32	43.12	72.59	33.32	78.77	70.84	74.15	75.47	48.00	59.31	85.41	74.04	71.56	81.52	55.47	51.86	66.40	65.10	10.3
One-stage Methods																								
R^3Det† [4]	R-50-FPN	800×800	61.35	41.79	71.48	81.57	34.63	73.38	77.85	62.43	26.93	76.69	76.54	37.12	51.95	78.96	65.28	61.17	81.50	53.31	41.65	64.32	61.02	13.2
RetinaNet-O [23]	R-50-FPN	800×800	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.20	69.25	79.54	32.14	44.87	77.71	67.57	61.09	81.46	47.33	38.01	60.24	57.55	7.0
S^2A-Net [11]	R-50-FPN	800×800	65.40	42.04	75.15	83.91	36.01	72.61	28.01	65.09	75.11	75.56	80.47	35.91	52.10	82.33	65.89	66.08	84.61	54.13	48.00	69.67	62.90	12.4
TCD [86]	R-50-FPN	800×800	40.38	85.89	67.95	65.07	81.61	74.74	56.00	48.72	73.48	45.30	83.55	78.22	79.33	57.22	78.03	33.60	70.54	68.99	48.46	63.79	65.04	-
Oriented RepPoints [57]	R-50-FPN	800×800	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	66.71	16.2
Baseline (FCOS-R)	R-50-FPN	800×800	60.42	39.62	72.67	87.71	35.13	73.26	80.24	61.14	28.38	77.95	80.39	45.61	53.08	86.75	68.35	68.31	81.79	55.59	45.33	65.76	63.37	19.1
Ours (FCOS-R)	R-50-FPN	800×800	68.07	50.29	71.68	86.45	39.89	79.66	84.25	70.64	34.93	77.97	79.30	46.09	55.97	86.96	70.52	68.69	81.23	56.49	50.35	65.76	66.17	14.1
Ours (Oriented RepPoints)	R-50-FPN	800×800	70.54	51.14	78.51	88.06	40.29	79.99	82.22	72.03	39.42	80.63	81.25	47.74	57.27	88.41	77.39	68.30	88.10	58.50	51.29	69.62	68.54	14.1
Ours (Oriented RepPoints)	R-101-FPN	800×800	71.24	54.28	79.08	87.68	41.96	80.66	84.29	73.27	38.78	79.85	79.95	48.15	57.15	87.47	75.01	66.66	87.48	57.72	50.63	70.61	68.60	12.9
Ours (Oriented RepPoints)	ConvNeXt-T-FPN	800×800	75.08	44.46	79.92	88.70	40.89	80.54	86.13	72.93	38.84	78.43	82.72	45.56	57.79	87.42	81.01	70.07	87.68	60.23	49.77	68.26	68.82	10.4
Ours (Oriented RepPoints)	Swin-T-FPN	800×800	75.08	54.45	80.13	86.44	41.28	80.47	86.38	76.35	40.12	80.77	82.49	47.99	58.03	87.44	80.65	74.45	88.71	65.89	50.87	68.17	70.31	13.7

TABLE VI

COMPARISON RESULTS WITH RELATED METHODS ON THE DOTA-1.0 TEST SET. EACH COLUMN'S BEST AND SECOND-BEST RESULTS ARE INDICATED BY RED AND BLUE COLORS, RESPECTIVELY. R-50-FPN REPRESENTS RESNET50-FPN [67]. R-101 REPRESENTS RESNET101. R-101-FPN REPRESENTS RESNET101-FPN. RER-50-FPN REPRESENTS RERESNET50-FPN [84]. D-53 REPRESENTS DARKNET53 [28]. DLA-34 REPRESENTS [87]. H-104 REPRESENTS HOURGLASS104 [88]. THE RESULTS OF RELATED METHODS ARE COLLECTED FROM ORIGINAL PAPERS

Methods	Backbone	Image Size	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP(%)
Two-stage Methods																		
Faster RCNN-O [17]	R-101	1024×1024	79.09	69.12	17.17	63.49	34.20	37.16	36.20	89.19	69.60	58.96	49.40	52.52	46.69	44.80	46.30	52.93
R-DFPN [89]	R-101-FPN	800×800	80.92	65.82	33.77	58.94	55.77	50.94	54.78	90.33	66.34	68.66	48.73	51.76	55.10	51.32	35.88	57.94
R^2CNN [90]	R-101	800×800	80.94	65.67	35.34	67.44	59.92	50.91	55.81	90.67	66.92	72.39	55.06	52.23	55.14	53.35	48.22	60.67
RRPN [91]	R-101	800×800	88.52	71.20	31.66	59.30	51.85	56.19	57.25	90.81	72.84	67.38	56.69	52.84	53.08	51.94	53.58	61.01
ICN [92]	R-101-FPN	-	81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23	68.16
CSL [12]	R-50-FPN	800×800	89.09	79.22	45.78	67.94	66.58	56.29	71.55	90.80	80.50	76.61	58.53	61.01	59.92	69.51	57.59	68.73
RoI-Transformer [2]	R-101-FPN	1024×1024	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
CAD-Net [3]	R-101-FPN	1600×1600	87.80	82.40	49.40	73.50	71.10	63.50	76.70	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
SCRDet [43]	R-50-FPN	800×800	89.48	79.83	50.43	62.51	76.72	74.31	85.42	90.86	84.19	82.51	42.84	60.15	74.18	70.62	51.81	71.72
FR-est [55]	R-101-FPN	1000×1000	89.63	81.17	50.44	70.19	73.52	77.98	86.44	90.82	84.13	83.56	60.64	66.59	70.59	66.72	60.55	74.20
Gliding Vertex [5]	R-101-FPN	1024×1024	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
DODet [41]	R-50-FPN	1024×1024	89.34	84.31	51.39	71.04	79.04	82.86	88.15	90.90	86.88	84.91	62.69	67.63	75.47	72.22	45.54	75.49
ReDet [84]	ReR-50-FPN	1024×1024	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
QPDet [46]	R-50-FPN	1024×1024	89.55	83.66	54.06	73.93	78.93	83.08	88.29	90.89	86.60	84.80	62.03	65.55	74.16	70.09	58.16	76.25
Oriented R-CNN [40]	R-101-FPN	1024×1024	88.86	83.48	55.27	76.92	74.27	82.10	87.52	90.90	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28
OSKDet [56]	R-101-FPN	1024×1024	90.07	87.08	54.16	75.61	72.64	76.86	87.63	90.77	79.10	86.88	59.88	71.28	75.16	71.71	66.67	76.37
One-stage Methods																		
PIoU [93]	DLA-34	512×512	80.90	69.70	24.10	60.20	38.30	64.40	64.80	90.90	77.20	70.40	46.50	37.10	57.10	61.90	64.00	60.50
DRN [42]	H-104	1024×1024	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
O^2-DNet [54]	H-104	511×511	89.31	82.14	47.33	61.21	71.32	74.03	78.62	90.76	82.23	81.36	60.93	60.17	58.21	66.98	61.03	71.04
BBAVectors [45]	R-101	608×608	88.35	79.96	50.69	62.18	78.43	78.98	87.94	90.85	83.58	84.35	54.13	60.24	65.22	64.28	55.70	72.32
CFC-Net [64]	R-50-FPN	800×800	89.08	80.41	52.41	70.02	70.02	78.11	87.21	90.89	84.47	85.64	60.51	61.52	67.82	68.02	50.09	73.50
R^3Det [4]	R-101-FPN	800×800	88.76	83.09	50.91	67.27	76.23	80.39	86.72	90.78	84.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79
S^2A-Net [11]	R-50-FPN	1024×1024	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
Oriented RepPoints [57]	R-50-FPN	1024×1024	87.02	83.17	54.13	71.16	80.18	78.40	87.28	90.90	85.97	86.25	59.90	70.49	73.53	72.27	58.97	75.97
R^3Det-GWD [14]	R-50-FPN	800×800	88.82	82.94	55.63	72.75	78.52	83.10	87.46	90.21	86.36	85.44	64.70	61.41	73.46	76.94	57.38	76.34
GGHL [15]	D-53	800×800	89.74	85.63	44.50	77.48	76.72	80.45	86.16	90.83	88.18	86.25	67.07	69.40	73.38	68.45	70.14	76.95
DAL [49]	R-50-FPN	800×800	89.69	83.11	55.03	71.00	78.30	81.90	88.46	90.89	84.97	87.46	64.41	65.65	76.86	72.09	64.35	76.95
Baseline	R-50-FPN	1024×1024	87.63	78.81	47.29	71.69	80.59	80.20	88.05	90.30	82.02	80.63	62.70	64.96	76.68	73.06	71.63	75.75
Ours	R-50-FPN	1024×1024	87.96	79.45	52.98	73.12	80.07	80.34	88.90	90.89	82.97	84.60	64.40	61.88	77.95	80.09	72.15	77.18

decoupled heterogeneous structure. Moreover, the performance can be further promoted by 0.64% mAP after introducing the JQE module. The results suggest that the adaptive aggregation of classification and regression information helps to estimate more accurate localization quality. Besides, introducing JQE with ISM and PBL, respectively, can achieve 0.47% mAP and 0.58% mAP gains, verifying the complementary effect of these components.

In Fig. 6, we visualize the detection results of baseline FCOS-R and the proposed TSH. It can be seen that our TSH produces more accurate detections for oriented objects, such as objects in the three and four columns, owing to the introduced PBL. Besides, some wrongly detected instances could be avoided by our methods, such as the false positives in the first column and the misclassified object in the second column. The results show the advantages of the proposed ISM.

2) Effectiveness of ISM: We further analyze the effectiveness of each submodule in ISM. As shown in Table II, the SA module improves the mAP by 0.35%, which enhances the feature response of objects while suppressing the background noise. When adding the IC loss directly on the feature map, the model is hard to converge. While adding the interclass CD alone, there is no obvious change in result compared with the baseline. Then, we integrate these two components, and the model can achieve 0.58% mAP improvement over the baseline, showing their mutual positive effectiveness. This is because interclass CD performs channel swapping to make the feature channels class-aligned. Thus, the IC loss can impose supervision on the classwise features to suppress the

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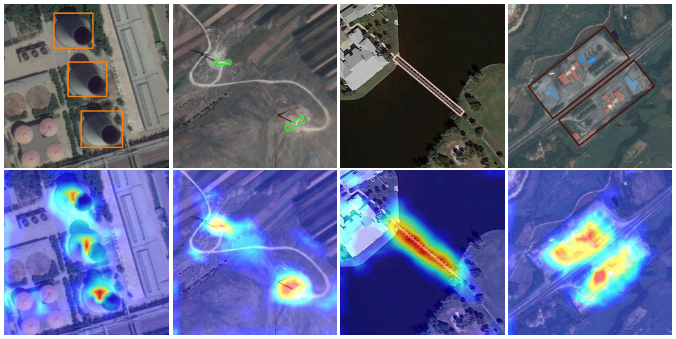


Fig. 7. Visualization of feature maps after applying the SA module. The top row is the ground truth, and the bottom row is the enhanced features.

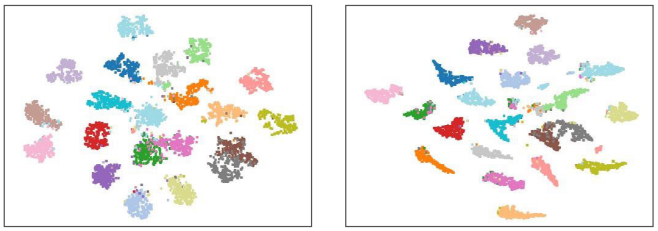


Fig. 8. t-SNE visualization of feature distributions of different classes before and after applying the interclass CD and IC loss.

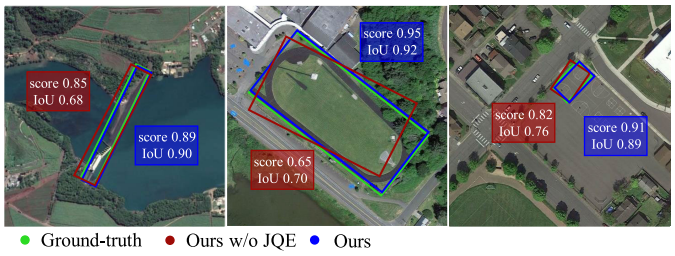


Fig. 9. Effect of JQE module on detection results.

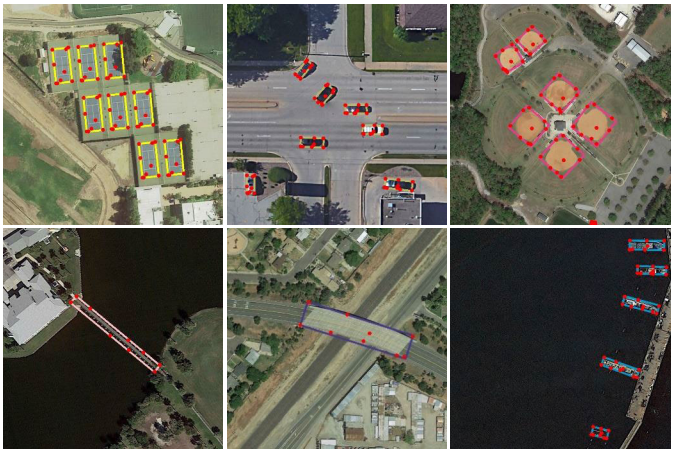


Fig. 10. Visualization of learned points and converted oriented boxes.

inference more effectively. Moreover, we compare our ISM with related feature enhancement strategies for interference suppression, as shown in Table III. The results show that the proposed ISM achieves the most significant improvement over the comparison methods, demonstrating the effectiveness of our tailor-designed structure.

Fig. 7 visualizes the enhanced feature map after applying the SA module. It can be observed that the feature maps accurately cover the foreground objects; thus, the network can concentrate more on the informative object regions. In addition, we visualize the feature distributions of different classes through t-distributed stochastic neighbor embedding (t-SNE) in Fig. 8. After adding the CD and IC loss, the feature distribution of each class is more compact and separated from other classes.

3) Effectiveness of JQE: To analyze the effectiveness of the proposed JQE module, we present an ablation study in Table II. We first replace the QFL loss in baseline method with original FL, and the performance drops from 63.37% mAP to 62.57%. This suggests the effectiveness of localization quality estimation. Then, we use a simple concatenation followed by a 3×3 convolution to achieve the feature interaction (FI) between classification and regression. This brings an improvement of 0.38% mAP over the baseline. Also, the results of applying our adaptive fusion module (AFM) are higher than using the simple combination, demonstrating that our AFM is more effective in adaptively aggregating the classification and localization information. In addition, we visualize some examples of the detection results in Fig. 9. Our method equipped with the JQE module estimates more reliable localization quality, thus maintaining more accurate predictions of high IoU with the ground truth.

4) Effectiveness of PBL: As shown in Table II, we examine the importance of each submodule in PBL. When applying the point representation (PR) and deformable convolution (DeformConv) [94] to aggregate the information of dispersible points, the performance is improved by 0.82% mAP over the baseline, validating that the localization information provided by adaptive points is effective to locate the aerial objects. Then, we replace the DeformConv with the proposed DIA module, and the performance can be further boosted by 0.83% mAP. This is because the DIA module enhances the feature representation of these points to enclose the object by exploiting the relationships between the dispersible points. Besides, we show some examples of the learned points in Fig. 10. It can be observed that the adaptive points are distributed accurately over the objects and well capture the structures of objects with arbitrary shapes and orientations. With the help of this point-based learning, our method can perform accurate localization.

5) Hyperparameter Analysis: In Table IV, we examine the influence of loss weight factor λ in the proposed ISM, which is used to adjust the IC. As λ increases, the influence of interference suppression is stronger, and our method achieves the best performance at $\lambda = 0.5$. When λ becomes larger, the performance drops slightly. The possible reason is that excessive interference suppression may also affect the representation ability of features. Finally, we select $\lambda = 0.5$ in our experiments.

C. Comparison With State-of-the-Art Methods

Based on four publicly available aerial object detection datasets, including DIOR-R, DOTA-1.0, HRSC2016, and

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Fig. 11. Comparison of different methods on the DIOR-R dataset. Our method produces more accurate detections and reduces false positives.



Fig. 12. Qualitative detection results of our TSH on (a) DIOR-R, (b) DOTA-1.0, (c) HRSC2016, and (d) UCAS-AOD datasets, respectively. Boxes with different colors denote objects of different categories.

UCAS-AOD, the proposed method is compared with other state-of-the-art methods.

1) Results on DIOR-R: In Table V, we compare our TSH with related methods on DIOR-R dataset. Our TSH achieves an mAP of 66.17% based on R-50-FPN. Compared with the baseline FCOS-R, TSH achieves an improvement of 2.8% mAP. When compared with two-stage methods, the proposed TSH outperforms the advanced DODet [41] with 1.07% mAP. Moreover, we integrate our TSH into state-of-the-art one-stage method oriented RepPoints [57], which employs better label assignment strategy. It can be seen that our method further improves the performance by 1.83%, reaching the mAP of 68.54%, achieving the best results among all compared methods. This verifies the effectiveness of our method on

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TABLE VII

COMPARISON RESULTS WITH RELATED METHODS ON THE HRSC2016 DATASET. R-50-FPN REPRESENTS RESNET50-FPN. R-101-FPN REPRESENTS RESNET101-FPN. RER-50-FPN REPRESENTS RERESNET50-FPN [84]. H-104 REPRESENTS HOURGLASS104 [88]. THE RESULTS OF RELATED METHODS ARE COLLECTED FROM ORIGINAL PAPERS

Methods	Backbone	Image Size	mAP(07)	mAP(12)
Two-stage Methods				
R^2CNN [90]	R-101	800×800	73.07	79.73
RRPN [91]	R-101	800×800	79.08	85.64
RoI-Transformer [2]	R-101-FPN	512×800	86.20	-
Gliding Vertex [5]	R-101-FPN	512×800	88.20	-
CSL [12]	R-50-FPN	800×800	89.62	96.10
FR-est [55]	R-101-FPN	-	89.70	-
AOPG [73]	R-50-FPN	800×1333	90.34	96.22
Oriented R-CNN [40]	R-50-FPN	800×800	90.40	96.50
ReDet [84]	ReR-50-FPN	512×800	90.46	97.63
QPDet [46]	R-50-FPN	800×1333	90.47	96.60
CGCDet [85]	R-50-FPN	-	90.57	**97.86**
DODet [41]	R-101-FPN	800×1333	**90.89**	97.14
One-stage Methods				
DRN [42]	H-104	768×768	-	92.70
CenterMap-Net [95]	R-50-FPN	-	-	92.80
R^3Det [4]	R-101-FPN	800×800	89.26	96.01
CFC-Net [64]	R-101-FPN	800×800	89.50	-
RetinaNet-O [23]	R-101-FPN	800×800	89.18	95.21
DAL [49]	R-101-FPN	800×800	89.77	-
S^2A-Net [11]	R-101-FPN	512×800	90.17	95.01
Oriented RepPoints [57]	R-50-FPN	800×800	90.38	97.26
TCD [86]	R-50-FPN	-	90.60	-
Ours	R-50-FPN	800×800	90.07	97.31

object detection in aerial images. In addition, we evaluate the performance of our method with more advanced backbones. Table V shows that our method achieves outstanding performance with different backbones. In this work, we choose R-50-FPN as backbone considering the trade-off of efficiency and accuracy as well as making a fair comparison with previous methods. Fig. 11 shows some qualitative results of the comparison methods on the DIOR-R dataset. We see that our TSH can perform accurate localization and effectively avoid false positives.

2) Results on DOTA-1.0: The experimental results of different methods on the DOTA-1.0 dataset are described in Table VI. In our experiments, we do not leverage multiscale data augmentation and a large backbone network during training. We see that our TSH yields the mAP of 77.18%, which consistently outperforms the baseline and surpasses the state-of-the-art methods. Fig. 12(b) visualizes some detection results from the DOTA-1.0 dataset of our TSH.

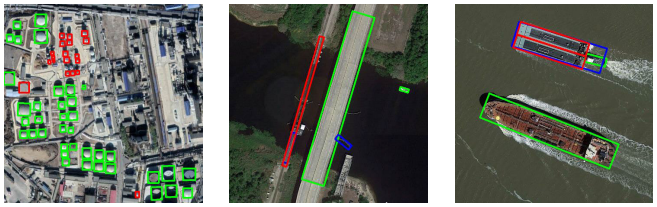
3) Results on HRSC2016: Ship objects in the HRSC2016 dataset show arbitrary orientations and large aspect ratios, making it challenging for oriented angle regression. In Table VII, our method yields 90.07% mAP when using VOC07 [77] metric and 97.31% mAP when using VOC12 [78] metric. The results is superior to the existing methods on HRSC2016, validating that our method is effective for detecting objects with large aspect ratios. Some qualitative results from HRSC2016 of our TSH are illustrated in Fig. 12(c), further verifying the excellent performance of our method.

4) Results on UCAS-AOD: We conduct comparison experiments on UCAS-AOD, depicted in Table VIII. Our TSH achieves the mAP of 90.20%, which is superior to previous methods. Fig. 12(d) is some qualitative results from UCAS-AOD of the proposed method. We see that our TSH can estimate accurate orientation of the square-like airplanes, verifying the effectiveness of our method.

TABLE VIII

COMPARISON RESULTS WITH RELATED METHODS ON THE UCAS-AOD DATASET. THE RESULTS OF RELATED METHODS ARE COLLECTED FROM ORIGINAL PAPERS, AND "†" DENOTES OUR REPRODUCED RESULT

Methods	Image Size	car	airplane	mAP(%)
Two-stage Methods				
Faster RCNN-O [17]	800×800	86.87	89.86	88.36
RoI-Transformer [2]	800×800	88.02	90.02	89.02
CSL† [12]	800×800	88.28	90.21	89.25
Gliding Vertex† [5]	800×800	88.40	90.13	89.26
DODet† [41]	800×800	88.25	90.39	89.32
QPDet† [46]	800×800	88.65	90.35	89.50
AOPG† [73]	800×800	88.74	90.34	89.54
Oriented R-CNN† [40]	800×800	89.07	90.35	89.71
ReDet† [84]	800×800	89.47	90.50	89.99
One-stage Methods				
CFC-Net [64]	800×800	89.29	88.69	89.49
R^3Det† [4]	800×800	88.70	90.35	89.52
RetinaNet-O [23]	800×800	84.64	90.51	87.57
DAL [49]	800×800	89.25	90.49	89.87
S^2A-Net† [11]	800×800	89.44	90.34	89.89
Oriented RepPoints [57]	800×800	89.51	90.70	90.11
Ours	800×800	**89.66**	**90.74**	**90.20**



(a) (b) (c)

Fig. 13. Some examples of failure cases. Green boxes indicate correct results, while red and blue boxes indicate false negatives and false positives, respectively. (a) Small objects. (b) Objects with extreme aspect ratios. (c) Densely arranged objects.

D. Analysis of Failure Cases

We show some examples of cases where our method does not perform well in Fig. 13. Although our method achieves superior performance, there are still some false detection results. For example, in Fig. 13(a), our method fails to detect some small objects, as the network cannot extract effective features for these objects with limited sizes. In Fig. 13(b), the insufficient receptive fields of features incur missing detection of objects with extreme aspect ratios. Besides, it is difficult to distinguish densely arranged objects, resulting in some inaccurate detections, as shown in Fig. 13(c). These cases are still very challenging, and we will further research on them in our future work.

V. CONCLUSION

In this article, we propose an effective aerial object detection framework, i.e., TSH network. We first employ the proposed ISM to suppress both the background and interclass interference in aerial images and obtain more discriminative features. Then, to predict more reliable localization confidence, we utilize the JQE module to adaptively combine the enhanced classification and regression features. For the regression, we design the PBL to accurately locate the arbitrary-oriented objects with a set of learnable points. Our method utilizes different structures for classification and regression tasks, according to the characteristics of aerial images. Extensive experiments demonstrate that our TSH achieves state-of-the-art performance on several aerial detection datasets.

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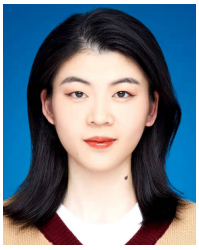
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